

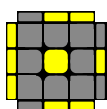
OLL 1

3x3 - OLL - Dot Case

setup:

 $F R' F' R U2' F R' F' R2' U2' R'$

Standard Alg:

 $(R U2 R') (R' F R F') U2 (R' F R F')$ ✓ $R U2 R2 F R F' U2 R' F R F'$ ✓ $y R U' R2 D' r U' r' D R2 U R'$ ✓ $f R U R' U' R f' U' r' U' R U M'$ ✓ $L' U2 L2 F' L' F U2 L F' L' F$ 

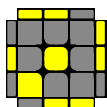
OLL 2

3x3 - OLL - Dot Case

setup:

 $f U R U' R' f' F U R U' R' F'$

Standard Alg:

 $F (R U R' U') F' f' (R U R' U') f'$ ✓ $y' R U' R2 D' r U' r' D R2 U R'$ ✓ $F R U R' U' S R U R' U' f'$ ✓ $F R U R' U' F' f' R U R' U' f'$ ✓ $y r U r' U2 R U2 R' U2 r U' r'$ 

OLL 3

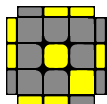
3x3 - OLL - Dot Case

setup:

 $F U R U' R' F' U f' U R U' R' f' y$

Standard Alg:

 $y' f' (R U R' U') f' (U') F (R U R' U') F'$ ✓ $y' f' R U R' U' f' U' F R U R' U' F'$ ✓ $y R' F2 R2 U2 R' F R U2 R2 F2 R$ ✓ $r' R2 U R' U' r U2 r' U M'$ ✓ $M R U R' U' r U2 r' U M'$ 



OLL 4

3x3 - OLL - Dot Case

setup:

FURUR'F'U'fURUR'f'y

Standard Alg:

y'f(RUR'U')f'(U)F(RUR'U')F'

✓ y'R'F2R2U2R'F'R U2R2F2R



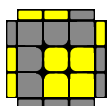
✓ y'fRUR'U'f'UfRUR'U'F'



✓ R'FRF'U'SR'U'RURS'



✓ yFURUR'F'U'FRUR'U'F'



OLL 5

3x3 - OLL - Square Shapes

setup:

r'U'RUR'R'U2'r

Standard Alg:

r'U2(RUR'U)r

✓ r'U2RUR'Ur



✓ y2l'U2LUL'Ul



✓ y2R'F2rUr'FR



✓ y2R'F2LFL'FR

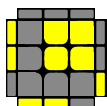


Visualize with Mind Maps

See how Mind Maps connect all the facts and figures you upload to NotebookLM



Google



OLL 6

3x3 - OLL - Square Shapes

setup:

rUR'URU2'r'

Standard Alg:

rU2(R'U'R U')r'

✓ rU2R'U'R U'r'



✓ FU'R2DR'U'R D'R2UF'

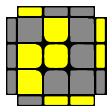


✓ y2lU2L'U'LU'l'



✓ y'x'DR2U'R'UR'D'x





OLL 7



3x3 - OLL - Lightning Shapes

setup:

r U2' R' U' R U' r'

Standard Alg:

r (U R' U R) U2 r'

▼ r U R' U R U2 r'



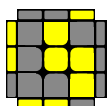
▼ S' R U R' U R U2 R' U S



▼ L' U2 L U2 L F' L' F



▼ y2 l U L' U L U2 l'



OLL 8



3x3 - OLL - Lightning Shapes

setup:

r' U2' R U R' U r y2'

Standard Alg:

y2 r' (U' R U' R') U2 r

▼ y2 r' U' R U' R' U2 r



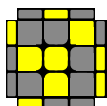
▼ l' U' L U' L' U2 l



▼ R U2 R' U2 R' F R F'



▼ R' F' r U' r' F2 R



OLL 9



3x3 - OLL - Fish Shapes

setup:

F U R U' R2' F' R U R U' R' y'

Standard Alg:

y (R U R' U') (R' F R) (R U R' U') F'

▼ y R U R' U' R' F R2 U R' U' F'



▼ R U2 R' U' S' R U' R' S

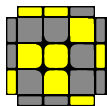


▼ y2 F' U' F r U' r' U r U r'



▼ y' L' U' L U' L F' L' F L' U2 L





OLL 10



3x3 - OLL - Fish Shapes

setup:

R U2' R' F R' F' R U' R U' R'

Standard Alg:

(R U R' U) (R' F R F') (R U2 R')

✓ R U R' U R' F R F' R U2 R'



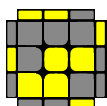
✓ y F U F' R' F R U' R' F' R



✓ y M' R' U2 R U R' U R U M



✓ y2 L' U' L U L F L2 U' L U F



OLL 11



3x3 - OLL - Lightning Shapes

setup:

M U' R U2' R' U' R U' R2' r

Standard Alg:

M (R U R' U R U2 R') U M'

✓ r' R2 U R' U R U2 R' U M'



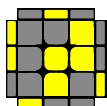
✓ y2 r U R' U R' F R F' R U2 r'



✓ S R U R' U R U2 R' U2 S'



✓ M R U R' U R U2 R' U M'



OLL 12



3x3 - OLL - Lightning Shapes

setup:

F U R U' R' F' U' F U R U' R' F'

Standard Alg:

y' M' (R' U' R U' R' U2 R) U' M

✓ y' M' R' U' R U' R' U2 R U' M



✓ F R U R' U' F' U F R U R' U' F'

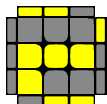


✓ y' S R' U' R U' R' U2 R U2 S'



✓ y l L2 U' L U' L' U2 L U' M'





OLL 13



3x3 - OLL - Knight Move Shapes

setup:

F' U' F r U' r' U r U r'

Standard Alg:

(r U' r) U' (r U r) (F' U F)

▼ F U R U 2 R' U' R U R' F'



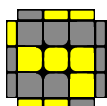
▼ F U R U' R 2 F' R U R U' R'



▼ r U' r' U' r U r' F' U F



▼ r U' r' U' r U r' y L' U L



OLL 14



3x3 - OLL - Knight Move Shapes

setup:

F U F' R' F R U' R' F' R

Standard Alg:

R' F (R U R') F' R (F U' F')

▼ R' F R U R' F' R F U' F'



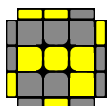
▼ r U R' U' r' F R 2 U R' U' F'



▼ F' U' L' U L 2 F L' U' L' U L



▼ l' U l U l' U' l F U' F'



OLL 15



3x3 - OLL - Knight Move Shapes

setup:

r' U' r U' R' U R r' U r

Standard Alg:

(r' U' r) (R' U' R U) (r' U r)

▼ r' U' r R' U' R U r' U r



▼ y 2 l' U' l l' U' L U l' U l

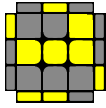


▼ r' U' M' U' R U r' U r



▼ y 2 R' F' R L' U' L U R' F R





OLL 16



3x3 - OLL - Knight Move Shapes

setup:

r U r' U R U' R' r U' r'

Standard Alg:

(r U r') (R U R' U') (r U' r')

✓ r U r' R U R' U' r U' r'



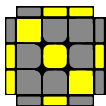
✓ r U M U R' U' r U' r'



✓ y2 R' F R U R' U' F' R U' R' U2 R



✓ y2 l U l' L U L' U' l U' l'



OLL 17



3x3 - OLL - Dot Case

setup:

F R' F' R U2' F R' F' R U' R U' R'

Standard Alg:

(R U R' U) (R' F R F') U2 (R' F R F')

✓ R U R' U R' F R F' U2 R' F R F'



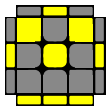
✓ y2 F R' F' R U S' R U' R' S



✓ y2 F R' F' R2 r' U R U' R' U' M'



✓ y' F' r U r' U' S' r' F r S'



OLL 18



3x3 - OLL - Dot Case

setup:

r' U2' R U R' U r2' U2' R' U' R U' r'

Standard Alg:

y (R U2 R') (R' F R F') U2 M' (U R U' r')

✓ y R U2 R2 F R F' U2 M' U R U' r'



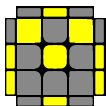
✓ r U R' U R U2 r2 U' R U' R' U2 r



✓ y F S' R U' R' S R U2 R' U' F'



✓ R D r' U' r D' R' U' R2 F R F' R



OLL 19



3x3 - OLL - Dot Case

setup:

F R' F' R M U R U' R' U' M'

Standard Alg:

M U (R U R' U') M' (R' F R F')

✓ y S' R U R' S U' R' F R F'



✓ M U R U R' U' M' R' F R F'

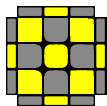


✓ R' U2 F R U R' U' F2 U2 F R



✓ r' R U R U R' U' r R2 F R F'





OLL 20

3x3 - OLL - Dot Case

setup:

r U R' U' M2' U R U' R' U' M'

Standard Alg:

(r U R' U') M2 (U R U' R') U' M'

✓ r U R' U' M2 U R U' R' U' M'



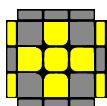
✓ M' U2 M U2 M' U M U2 M' U2 M



✓ S' R U R' S U' M' U R U' r'



✓ S R' U' R U R U R U' R' S'



OLL 21

3x3 - OLL - OCLL

setup:

R U R' U R U' R' U R U2' R' y'

Standard Alg:

(R U R' U) (R U' R' U) (R U2 R')

✓ R U R' U R U' R' U R U2 R'



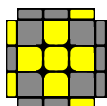
✓ y R U2 R' U' R U R' U' R U' R'



✓ y F R U R' U' R U R' U' R U R' U' F'



✓ R' U' R U' R' U R U' R' U2 R



OLL 22

3x3 - OLL - OCLL

setup:

R' U2' R2' U R2' U R2' U2' R'

Standard Alg:

R U2 (R2' U') (R2 U') (R2' U') U' R

✓ R U2 R2 U' R2 U' R2 U2 R



✓ R' U2 R2 U R2 U R2 U2 R'

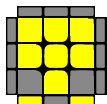


✓ f R U R' U' S' R U R' U' F'



✓ f R U R' U' f' F R U R' U' F'



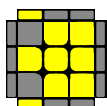


OLL 23 
3x3 - OLL - OCLL

setup:
R U2' R D R' U2' R D' R2'

Standard Alg:
R2 D (R' U2 R) D' (R' U2 R')

✓ R2 D R' U2 R D' R' U2 R'	 
✓ y2 R2 D' R U2 R' D R U2 R	 
✓ R U R' U R U2 R2 U' R U' R' U2 R	 
✓ y R U R' U' R U' R' U2 R U' R' U2 R U R'	 

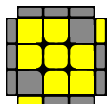


OLL 24 
3x3 - OLL - OCLL

setup:
F R' F' r U R U' r'

Standard Alg:
(r U R' U') (r' F R F')

✓ r U R' U' r' F R F'	 
✓ y2 R' F' r U R U' r' F	 
✓ y' x' R U R' D R U' R' D' x	 
✓ y R U R D R' U' R D' R2	 

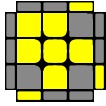


OLL 25 
3x3 - OLL - OCLL

setup:
R' F' r U R U' r' F y'

Standard Alg:
y (F' r U R') (U' r' F R)

✓ R U2 R D R' U2 R D' R2	 
✓ y F' r U R' U' r' F R	 
✓ F R' F' r U R U' r'	 
✓ x R' U R D' R' U' R D x'	 

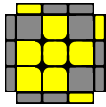


OLL 26 
3x3 - OLL - OCLL

setup:
R U R' U R U2' R' y'

Standard Alg:
y R U2 (R' U' R U) R'

✓ y R U2 R' U' R U' R'	 
✓ R' U' R U' R' U2 R	 
✓ y2 L' U' L U' L' U2 L	 
✓ y2 L' U R U' L U R'	 

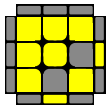


OLL 27 
3x3 - OLL - OCLL

setup:
R U2' R' U' R U' R'

Standard Alg:
(R U R' U) (R U2 R')

✓ R U R' U R U2 R'	 
✓ y' R' U2 R U R' U R	 
✓ y L' U2 L U L' U L	 
✓ y2 L U L' U L U2 L'	 

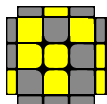


OLL 28 
3x3 - OLL - All Corners Oriented

setup:
R U R' U' M' U R U' r'

Standard Alg:
(r U R' U') M (U R U' R')

✓ r U R' U' M U R U' R'	 
✓ R' F R S R' F' R S'	 
✓ r U R' U' r' R U R U' R'	 
✓ y2 M' U M U2 M' U M	 



OLL 29

3x3 - OLL - Awkward Shapes

setup:

M F R' F' R U R U' R' U' M'

Standard Alg:

y (R U R') U' (R U' R') (F' U' F) (R U R')

✓ r2 D' r U r' D r2 U' r' U' r



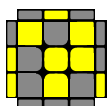
✓ y R U R' U' R U' R' F' U' F R U R'



✓ y S' R U R' U' R' F R F' U S



✓ M U R U R' U' R' F R F' M'



OLL 30

3x3 - OLL - Awkward Shapes

setup:

F U R U2' R' U R U2' R' U' F' y2'

Standard Alg:

y2 F U (R U2 R') U' (R U2 R') U' F'

✓ y' r' D' r U' r' D r2 U' r' U r U' r'



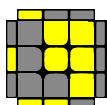
✓ y2 F U R U2 R' U' R U2 R' U' F'



✓ y2 F R' F R2 U' R' U' R U R' F2



✓ y S' R' U' R f R' U R U' F'



OLL 31

3x3 - OLL - P Shapes

setup:

R' F R U R' U' F' U R

Standard Alg:

(R' U' F) (U R U' R') F' R

✓ R' U' F U R U' R' F' R



✓ y2 S' L' U' L U L F' L' f

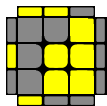


✓ y S R U R' U' f U' F



✓ y' F R' F' R U R U R' U' R U' R'





OLL 32

3x3 - OLL - P Shapes

setup:

f R' F' R U R U' R' S'

Standard Alg:
S (R U R' U') (R' F R f')

✓ S R U R' U' R' F R f'



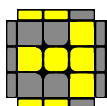
✓ y2 L U F' U' L' U L F L'



✓ R U B' U' R' U R B R'



✓ y' R' F R F' U' r U' r' U r U' r'



OLL 33

3x3 - OLL - T Shapes

setup:

F R' F' R U R U' R'

Standard Alg:
(R U R' U') (R' F R F')

✓ R U R' U' R' F R F'



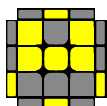
✓ y2 L' U' L U L F' L' F



✓ y2 r' F' r U r U' r' F



✓ R U R' F' U' F R U' R'



OLL 34

3x3 - OLL - C Shapes

setup:

F U R' U' R' F' R U R2' U' R' y2'

Standard Alg:
y2 R U R2 U' R' F (R U R U') F'

✓ y f R f' U' r' U' R U M'



✓ y2 R U R2 U' R' F R U R U' F'

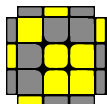


✓ F R U R' U' R' F' r U R U' r'



✓ y2 R U R' U' B' R' F R F' B





OLL 35



3x3 - OLL - Fish Shapes

setup:

R U2' R' F R' F' R2' U2' R'

Standard Alg:

(R U2 R') (R' F R F') (R U2 R')

✓ R U2 R2 F R F' R U2 R'



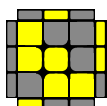
✓ f R U R' U' f' R U R' U R U2 R'



✓ y L' U2 L2 F' L' F L' U2 L



✓ R U2 R' d' R' F R U' R' F' R



OLL 36



3x3 - OLL - W Shapes

setup:

F' L F L' U' L' U' L U L' U L y2'

Standard Alg:

y2 (L' U' L U') (L' U L U) (L F' L' F)

✓ y R U R2 F' U' F U R2 U2 R'



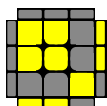
✓ y2 L' U' L U' L' U L U L F' L' F



✓ y2 R U R' F' R U R' U' R' F R U' R' F R F'



✓ y2 R' F' U' F2 U R U' R' F' R



OLL 37



3x3 - OLL - Fish Shapes

setup:

F R U' R' U R U R' F'

Standard Alg:

F R (U' R' U') (R U R') F'

✓ F R' F' R U R U' R'



✓ F R U' R' U' R U R' F'

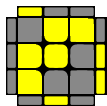


✓ y F' r U r' U' r' F r



✓ y2 r2 D' r U' r' D r U r





OLL 38

3x3 - OLL - W Shapes

setup:

FR'F'RURUR'U'R'UR'

Standard Alg:

(RUR'U)(RU'R'U')(R'FRF')

✓ RUR'URUR'R'UR'FRF'



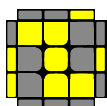
✓ yFRUR'SURUR'f'



✓ rUR'U'r'FRURUR'F'



✓ y2LUL'ULU'L'U'L'BLB'



OLL 39

3x3 - OLL - Lightning Shapes

setup:

LUF'U'L'ULFL'y'

Standard Alg:

yLF'(L'U'LU)FU'L'

✓ y'f'rUr'U'r'FrS



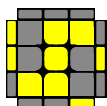
✓ y'RUR'F'U'FURU2R'



✓ yLF'L'U'LUFU'L'



✓ y'f'LFL'U'L'ULS



OLL 40

3x3 - OLL - Lightning Shapes

setup:

R'U'FURUR'R'FR'y'

Standard Alg:

yR'F(RUR'U')FUR

✓ yR'FRUR'U'FUR



✓ y'fR'F'FRURUR'S'

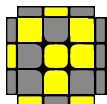


✓ RrDr'UrD'r'U'R'



✓ y'L'U'LFUF'U'L'U2L





OLL 41



3x3 - OLL - Awkward Shapes

setup:

F U R U' R' F' R U2' R' U' R U' R' y2'

Standard Alg:

y2 (R U R' U) (R U2 R) F (R U R' U) F'

✓ y2 R U R' U R U2 R' F R U R' U' F'



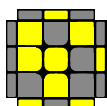
✓ y2 F U R2 D R' U' R D' R2 F'



✓ y' S U' R' F' U' F U R S'



✓ M U' F' L' U' L U F M'



OLL 42



3x3 - OLL - Awkward Shapes

setup:

F U R U' R' F' R' U2' R U R' U R

Standard Alg:

(R' U' R U') (R' U2 R) F (R U R' U) F'

✓ R' U' R U' R' U2 R F R U R' U' F'



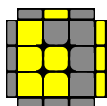
✓ y F S' R U R' U' F U S



✓ y R' F R F' R' F R F' R U R' U' R U R'



✓ y R' U' F2 u' R U R' D R2 B



OLL 43



3x3 - OLL - P Shapes

setup:

f' U' L' U L f

Standard Alg:

y R' U' (F' U F) R

✓ y R' U' F' U F R



✓ y2 F' U' L' U L F

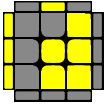


✓ f' L' U' L U f



✓ B' U' R' U R B





OLL 44

3x3 - OLL - P Shapes

setup:

f U R U' R' f'

Standard Alg:

f (R U R' U') f'

▼ f R U R' U' f'



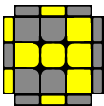
▼ y2 F U R U' R' F'



▼ y R U B U' B' R'



▼ y' L U F U' F' L'



OLL 45

3x3 - OLL - T Shapes

setup:

F U R U' R' F'

Standard Alg:

F (R U R' U') F'

▼ F R U R' U' F'



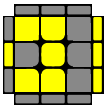
▼ y R' F' U' F U R



▼ y2 f U R U' R' f'



▼ y2 F' L' U' L U F



OLL 46

3x3 - OLL - C Shapes

setup:

R' U' F R' F' R U R

Standard Alg:

R' U' (R' F R F') U R

▼ R' U' R' F R F' U R



▼ R' F' U' F R U' R' U2 R

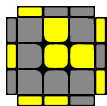


▼ y F R U R' U' F' U' R U R' U R U2 R'



▼ l' U2 L2 F' L' F U L' U l





OLL 47

3x3 - OLL - L Shapes

setup:

F' U' L' U L U' L' U L F

Standard Alg:

F' (L' U' L U) (L' U' L U) F

✓ y' F R' F' R U2 R U' R' U R U2 R'



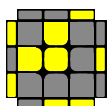
✓ F' L' U' L U L' U' L U F



✓ R' U' R' F R F' R' F R F' U R



✓ y' R' F' U' F U F' U' F U R



OLL 48

3x3 - OLL - L Shapes

setup:

F U R U' R' U R U' R' F'

Standard Alg:

F (R U R' U') (R U R' U') F'

✓ F R U R' U' R U R' U' F'



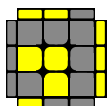
✓ y2 f U R U' R' U R U' R' f'



✓ R U2 R' U' R U R' U2 R' F R F'



✓ F R' F' U2 R U R' U R2 U2 R'



OLL 49

3x3 - OLL - L Shapes

setup:

r' U r2' U' r2' U' r2' U r' y2'

Standard Alg:

y2 r U' (r2 U) (r2 U) (r2) U' r

✓ y2 r U' r2 U r2 U r2 U' r



✓ l U' l2 U l2 U l2 U' l

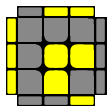


✓ R B' R2 F R2 B R2 F' R



✓ y2 R' F R' F' R2 U2 B' R B R'





OLL 50

3x3 - OLL - L Shapes

setup:

r U' r2' U r2' U r2' U' r

Standard Alg:

r' U (r2 U') (r2 U') (r2) U r'

✓ r' U r2 U' r2 U' r2 U r'



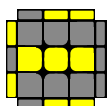
✓ y2 R' F R2 B' R2 F' R2 B R'



✓ y' R U2 R' U' R U' R' F R U R' U' F'



✓ y2 l' U l2 U' l2 U' l2 U l'



OLL 51

3x3 - OLL - Line Shapes

setup:

f U R U' R' U R U' R' f

Standard Alg:

f (R U R' U') (R U R' U') f'

✓ y2 F U R U' R' U R U' R' F'



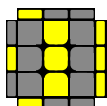
✓ f R U R' U' R U R' U' f'



✓ y' R' U' R' F R F' R U' R' U2 R



✓ y r' F' U' F U F' U' F U r



OLL 52

3x3 - OLL - Line Shapes

setup:

F R U R' d R' U' R U' R'

Standard Alg:

y2 R' (F' U' F U') (R U R' U) R

✓ y2 R' F' U' F U' R U R' U R



✓ R U R' U R U' B U' B' R'

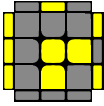


✓ R U R' U R d' R U' R' F'



✓ R U R' U R U' y R U' R' F'





OLL 53



3x3 - OLL - L Shapes

setup:

r' U2' R U R' U' R U R' U r

Standard Alg:

(r' U' R U') (R' U R U') (R' U2 r)

✓ r' U' R U' R' U R U' R' U2 r



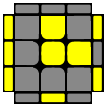
✓ y2 l' U' L U' L' U L U' L' U2 l



✓ y r' U2 R U R' U' R U R' U r



✓ y' l' U2 L U L' U' L U L' U l



OLL 54



3x3 - OLL - L Shapes

setup:

r U2' R' U' R U R' U' R U' r'

Standard Alg:

(r U R' U) (R U' R' U) (R U2 r')

✓ r U R' U R U' R' U R U2 r'



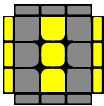
✓ y' r U2 R' U' R U R' U' R U' r'



✓ y' r U' R' U R U' R' U R U' R' U' r'



✓ y2 l U L' U L U' L' U L U2 l'



OLL 55



3x3 - OLL - Line Shapes

setup:

F R' F' U2' R U R' U R2' U2' R'

Standard Alg:

R U2 R2 (U' R U' R') U2 (F R F')

✓ y R' F U R U' R2 F' R2 U R' U' R



✓ y R' F R U R U' R2 F' R2 U' R' U R U R'

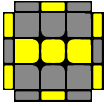


✓ R U2 R2 U' R U' R' U2 F R F'



✓ r U2 R2 F R F' U2 r' F R F'





OLL 56

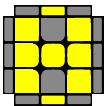


3x3 - OLL - Line Shapes

setup:

 $rU'rUR'U'RUR'U'rU'r'$

Standard Alg:

 $(rU'r)(UR'U'R')(UR'U'R')(rU'r)$ ✓ $rU'rUR'U'R'UR'U'R'rU'r'$ ✓ $rU'rUR'U'R'M'URU2r'$ ✓ $FRUR'U'RF'rUR'U'r'$ ✓ $r'U'rU'R'UR'U'R'UR'rUr$ 

OLL 57



3x3 - OLL - All Corners Oriented

setup:

 $rUR'U'MUR'U'R'$

Standard Alg:

 $(RUR'U')M'(UR'U'r')$ ✓ $RUR'U'M'UR'U'r'$ ✓ $yRU'R'S'RUR'S$ ✓ $yRU'R'S'RUR'S$ ✓ $RUR'U'R'rUR'U'r'$ SpeedCubeDB Created By Gil Zussman gil@speedcubedb.com